

thermodynamics - practice problems

name _____

1. A hot ceramic mug sits on a cool wooden table. Which way does heat flow, and when does it mostly stop?
2. A gas loses 30 J of heat to its surroundings while a piston does 80 J of work on it. What is the change in the gas's internal energy?
3. A gas loses 120 J of heat and does 40 J of work on its surroundings as it expands. What is the change in the gas's internal energy?
4. A gas receives 80 J of heat and expands for 2.0 s. What is the change in the gas's internal energy?
5. A system starts with 500 J of internal energy. During a process, 120 J of heat leaves the system and 80 J of work is done on it. What is its final internal energy?
6. In a refrigerator, the compressor does 200 J of work on the refrigerant gas while 0.50 kJ of heat leaves the gas through the coils on the back. What is the change in the gas's internal energy?

7. You rub your hands together, doing 30 J of work on them. During that time, 10 J of heat escapes from your hands into the air. How much does the internal energy of your hands change?

8. A 50 kg skateboarder drops into a half-pipe from rest at the top edge, 3.0 m above the bottom. They roll up the other side and stop 2.6 m above the bottom. How much energy left the skateboarder's system as heat?

9. A 58 g tennis ball is dropped from rest 2.0 m above the floor. It bounces back up to a height of 1.1 m. How much energy left the ball as heat during the bounce? What fraction of its energy did it lose?

10. A 30 kg sled starts from rest at the top of a hill 10 m tall and reaches the bottom moving at 12 m/s. How much energy left as heat? The slope is 50 m long. What was the average friction force?

11. A refrigerator makes the food inside colder and more orderly. Doesn't that decrease entropy and break the second law of thermodynamics?

12. A video shows spilled milk jumping up off the floor and back into a glass. What thermodynamics idea makes that look wrong?